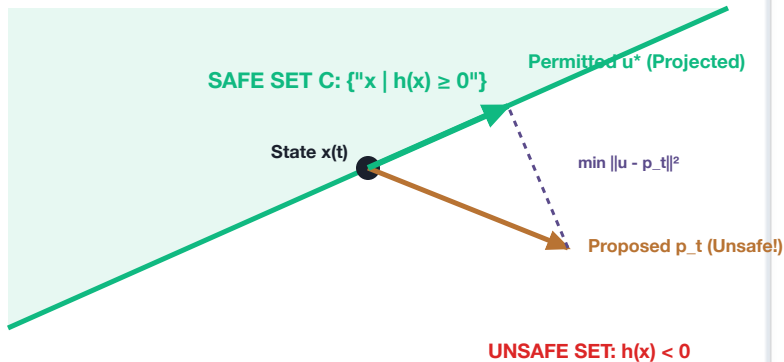


CONTROL BARRIER FUNCTION (CBF) MINIMAL-INTERVENTION PROJECTION

Quadratic Program Filter: $\min \|u - p_t\|^2$ subject to $L_f h(x) + L_g h(x) u + \alpha(h(x)) \geq 0$



THE BARRIER GUARANTEE

- **Minimal Intervention Principle:**

If proposed action p_t is safe, $u^* = p_t$ with zero distortion.
If p_t violates safety, QP applies minimal correction.

- **Forward Invariance (Nagumo's Theorem):**

If state starts in C ($h(x_0) \geq 0$), it remains inside C
for all future time $t \geq 0$: $h(x(t)) \geq 0$ certified.

- **Deterministic Active-Set QP Solver:**

Formulated with linear constraints in 3–6 variables.
Guaranteed convergence in ≤ 15 iterations ($\leq 175 \mu s$).

- **Zero Dynamic Allocation (malloc=0):**

Matrices pre-allocated in static SRAM.
Zero page faults or heap jitter in real-time loop.